

PROBLEM SET 1

(Due Sep 2, 2022 @ 12:00 pm in Gradescope and WileyPlus)

This assignment is to be completed individually. The number in parentheses next to each problem shows how many points the problem is worth in the overall assignment. Please show all work and use correct significant figures to receive full credit. Be sure to follow the problem formatting instructions to avoid unnecessary deductions. If you use an equation-solving tool (calculator, APEX, Solver, MATLAB), write out the equation(s) and note the tool that you used; otherwise, show all hand calculations used to solve any equations.

- (10 pts)** Read Chapter 1 from FR&B; which of the pursuits in Ch. 1 strikes you as the most suitable job environment for you, and why? Or is there something *not* listed that you plan to pursue? What would you consider your main strengths and interests that led you to pursue your choice of major? Your response should be typed and printed out, at least a half page one page max (12 pt font).
- (12 pts)** Using dimensional equations, convert
 - 2 weeks to microseconds
 - 38.1 ft/s to kilometers/h
 - $554 \text{ m}^4/(\text{day} \cdot \text{kg})$ to $\text{ft}^4/(\text{min} \cdot \text{lb}_m)$
- (10 pts)** Suppose you have a vessel containing 50 g of liquid toluene at 20 C.
 - What is the volume (mL) of toluene?
 - How many g-mols is it?
 - How many lb_m -mols is it?
 - Would you expect the SG of toluene to be greater than or less than that of water? Why?
- (6 pts)** A good engineer has a physical sense what a gram, kg, mg mean physically. Estimate (**one** significant figure) the mass for the following objects (you can use the Internet if necessary) in the units listed:
 - Dime - milligrams
 - I-Phone - grams
 - NC State Quarterback Devin Leary - kilograms
 - Tesla Model S - kilograms
- (6 pts)** A waste treatment pond is 50 m long, 25 m wide, and an average depth of 2 m. The density of the waste is $75.3 \text{ lb}_m/\text{ft}^3$. Calculate the weight of the pond contents in lb_f , using a single-dimensional equation for your calculation.
- (16 pts)** The density of a fluid is given by the empirical equation (below), where ρ is density (lb_m/ft^3) and P is pressure (lb_f/in^2).

$$\rho = 70.5 \exp(8.27 \times 10^{-7} P)$$

- (2 pts)** What are the units of 70.5 and 8.27×10^{-7} ?
- (6 pts)** Calculate the density in g/cm^3 for a pressure of $9.00 \times 10^6 \text{ N}/\text{m}^2$.
- (8 pts)** Derive a formula for ρ (g/cm^3) as a function of P (N/m^2). See Example 2.6-1 in book. Check your result using the solution from Part (b).

(40 pts) The remainder of the assignment (4 problems) will be completed online *individually* using WileyPLUS. *You do not have to turn in any paperwork with this portion of the assignment.* To access the assignment, first register using the link on the Moodle site, and then you can access the Assignment within

WileyPLUS. The due date for the WileyPLUS completion is the same as for the homework assignment. Note that the WileyPLUS assignment cannot be submitted late.

Honors Program Problem

The **Reynolds number** is a dimensionless group defined for a fluid flowing in a pipe, expressed as

$$Re = Dup/\mu$$

where D is pipe diameter, u is fluid velocity, ρ is fluid density, and μ is fluid viscosity. When the value of the Reynolds number is less than about 2100, the flow is *laminar*—that is, the fluid flows in smooth streamlines. For Reynolds numbers above 2100, the flow is *turbulent*, characterized by a great deal of agitation. Liquid methyl ethyl ketone (MEK) flows through a pipe with an inner diameter of 2.067 inches at an average velocity of 0.48 ft/s. At the fluid temperature of 20°C, the density of liquid MEK is 0.805 g/cm³ and the viscosity is 0.43 centipoise [1 cP = 1.00 x 10⁻³ kg/(m s)]. Without using a calculator, determine whether the flow is laminar or turbulent using estimation. Show your hand calculations.